

CHETAN AMRITANSHU

Backend Engineer | Distributed Systems | GenAI | Codeforces Expert

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github.com/ChetanAmritanshu | linkedin.com/in/chetan-amritanshu | Portfolio website

SUMMARY

Backend Engineer with 2.5+ years of experience building high-throughput Go services, distributed systems, and GenAI applications across trading and retail. Experienced in concurrency, event-driven architectures, Kafka/RabbitMQ, Redis coordination, fault-tolerant workflows, and low-latency backend systems.

EXPERIENCE

Oneture Technologies

Navi Mumbai

Associate Software Development Engineer | Backend / GenAI

Mar 2025 – Present

Trading Square-Off System – Go

- Engineered a high-throughput **Go backend** processing **5K+ trades/min** with **sub-120 ms P95 latency**.
- Architected **RabbitMQ-driven event workflows** with idempotent consumers, retries, and DLQs for failure-safe trade execution.
- Implemented **Redis-based distributed coordination and locking**, reducing database writes by **40%**.
- Built circuit breakers, timeouts, and fault-isolation strategies, contributing to **99.9%+ availability**.

AI-Powered Sales Advisor – Python / GenAI

- Built a microservice-based AI product discovery backend serving a catalog of **50K+ products**.
- Optimized **OpenSearch** indexing and retrieval for the 50K+ product catalog, reducing search latency by **30%**.

Newton School

Pune

Software Development Engineer + Instructor

Feb 2024 – Feb 2025

- Developed production backend services using **Spring Boot and Django**; optimized SQL queries and indexing, reducing API latency by **40%**.
- Mentored **100+ students** in Data Structures, Algorithms, and System Design.

PROJECTS

TicketForge – Failure-Driven High-Concurrency Ticketing System

github.com/ChetanAmritanshu/TicketForge

- Engineered a high-concurrency ticket allocation system handling **50,000 competing purchase attempts for 100 tickets** with **zero overselling**, durable ownership, and idempotent retries.
- Designed **PostgreSQL-backed transactional allocation**, durable request-ID idempotency, admission control, and bounded overload shedding while preserving correctness across multiple service instances.
- Achieved **3,066 req/s** with **427.9 ms P99** across three seller instances on a 50K-request validation, completing with **0 correctness violations** and exactly **100 unique ticket owners**.
- Engineered recovery for process restarts, PostgreSQL outages, slow-database degradation, connection-pool starvation, and post-commit response-loss ambiguity.

SKILLS

Languages: Go, C++, Java, Python, JavaScript/TypeScript, SQL

Backend: Gin, GORM, Spring Boot, Django, REST APIs, gRPC

Distributed Systems: Kafka, RabbitMQ, Redis, Concurrency, Event-Driven Architecture, Idempotency, Rate Limiting, Fault Isolation

Databases / Infra: PostgreSQL, MySQL, MongoDB, AWS EC2/S3, Docker, Linux, Git, Jenkins

EDUCATION

Indian Institute of Information Technology, Nagpur

2020 – 2024

B.Tech in Computer Science and Engineering

GPA: 7.84 / 10

ACHIEVEMENTS

- Codeforces Expert** (Max Rating: 1704) • **CodeChef 4-Star** • **Smart India Hackathon 2022 Winner**
- CodeChef Starters 255 – Rank 97 / 22K+ contestants**
- Educational Codeforces Round 188 – Rank 342 / 30K+ contestants**
- CodeChef Starters 253 – Rank 155 / 22K+ contestants**